

Practical FaaS Performance Variability Implications in the Serverless Image Optimization Scenario

Master's Thesis

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(Unterschrift) Piotr Witkowski, Berlin, 4. Januar 2025

*https://www.static.tu.berlin/fileadmin/www/10000060/FSC/Promotion__Habilitation/Dokumente/Grundsatzgute_wissenschaftliche_Praxis_2017.pdf

Abstract

Kurzfassung

Contents

1	Introduction	8
1.1	Objectives of the Thesis	8
1.2	Structure	9
2	Background	10
2.1	Imaging and Photography	10
2.1.1	Digital imaging	11
2.1.2	Image data compression	12
2.1.3	Image formats and their usage	12
2.1.4	Raster and vector graphics	12
2.2	Cloud Computing	12
2.2.1	Cloud deployment models	12
2.2.2	Event-driven architectures	12
2.2.3	Cloud service benchmarking	12
2.2.4	Performance variability of serverless platforms	12
2.3	Image Optimization for the Web	12
2.3.1	Impact of images on browsing experience	12
2.3.2	Bandwidth usage, visual quality, and browser support evolution	12
2.3.3	Optimizing latency of image delivery	12
2.3.4	Deployment methods for cloud-based image optimization	12
3	System Design	13
3.1	Image processing workload design	13
3.1.1	Serverless environment selection	13
3.1.2	Image manipulation library selection	13
3.1.3	Input image and output image selection	13
3.2	Benchmark design	13
3.2.1	Serverless platform and underlying hardware	13
3.2.2	Time of execution	13
3.2.3	Output formats and image widths	13
3.2.4	Available CPU / memory	13
3.2.5	Cold starts	13
3.3	Workload generation and benchmark execution	13
3.3.1	Serverless benchmark deployment	13
3.3.2	Scheduling workload executions	13
3.3.3	Results collection and data analysis	13
4	Results	14
4.1	Performance metrics	14
4.1.1	Cold start duration	14
4.1.2	Image processing duration	14
4.1.3	Memory utilization	14
4.2	Performance variability	14
4.2.1	General variability in performance metrics	14
4.2.2	Temporal variability	14
4.2.3	By image format	14
4.2.4	By platform configuration	14

5	Evaluation	15
5.1	Performance variability analysis	15
5.2	User vs operator perspective	15
5.3	Cloud computing as a utility	15
6	Related Work	16
6.1	Serverless performance research	16
6.2	Image processing research	16
7	Conclusion	17

1 Introduction

The digital revolution has fundamentally transformed the way we create, consume, and interact with visual information. Images are no longer only physical items around us, they are inherent components of our online experience, shaping perceptions and influencing decisions. As the Internet continues to evolve, so too grows the demand for rich and engaging visual content. This demand, however, comes at a cost. Images, while visually appealing, are significantly impacting website performance, not only by generating the highest number of requests, but also by consuming the most bytes during page load of an average site. Existing studies have shown a strong correlation between page load times and user engagement, conversion rates, and even search engine rankings. Minimizing image size is therefore fundamental for efficient bandwidth use and plays a crucial role in optimizing website performance.

Image optimization practices for the web have evolved over time, following developments of the state-of-the-art image formats and codecs. These solutions perform operations such as image resizing and format conversion, and may allow for storing optimized versions of original files for future use. While the optimization itself is a relatively specialized task, its deployment models follow familiar and established patterns in the general software industry: we can find them in software-as-a-service offerings, integrated into web hosting and content delivery platforms, but also as open-source solutions for self-hosting.

This last option, leveraging open-source image processing libraries, presents a compelling opportunity for our research. Unlike proprietary solutions, open-source projects offer the flexibility to be deployed across a variety of environments, granting builders granular control over the optimization process. This allows not only to measure influence of different libraries and optimization parameters, but also – and crucially for this thesis – enables realistic benchmarking of the underlying platforms on which these operations are performed. By employing these workloads, our research aims to accurately measure cloud service performance under the realistic conditions of real-world image optimization scenarios.

Given the event-driven nature of the optimization workload in response to requests for images, serverless cloud compute emerges as a well-suited deployment option for these systems. However, the varying demands for image processing, including different image sizes and diverse range of image formats present on the web, raise questions about the suitability and performance consistency of these platforms for the use-case. Furthermore, the emergence of function as a service (FaaS) platforms introduces a new dimension to cloud-based, self-hosted image optimization. FaaS offers a compelling execution environment for image processing tasks thanks to its on-demand scalability and event-driven nature. However, the performance variability inherent in FaaS platforms, attributed to diverse factors, may impact efficiency of the process and applicability of the environment for the use case.

1.1 Objectives of the Thesis

This thesis aims to investigate whether FaaS performance variability has measurable and meaningful influence for image optimization workloads. We will define set of representative metrics related to image optimization, reflecting realistic web usage. We will design and implement the optimization workflows for selected serverless environments. We will design and implement scheduling mechanism allowing us to simulate requests for optimized images. We will collect performance metrics across all parameters, apply statistical

methods for result comparisons, and visualize the results. We will investigate if more intensive optimization can be cost-efficient, considering optimization result caching, and content reuse for multiple viewers. We will provide recommendations that are within the scope of the benchmark, and answer the primary research question in the context of the scenario.

1.2 Structure

2 Background

This chapter provides readers with foundations for understanding the challenges and opportunities of serverless image processing. We begin by revisiting fundamental principles of imaging and photography. Subsequently, we describe core digital imaging concepts, emphasizing aspects relevant to web performance. We then examine the characteristics of serverless cloud computing, focusing on factors contributing to performance variability. Finally, we explore image optimization for the web, reviewing established techniques and the landscape of cloud-based solutions.

2.1 Imaging and Photography

Imaging is a broad term describing any process that helps us in visualizing and understanding physical objects, abstract concepts, or even sets of data. While this may be a purely philosophical activity, for the purpose of this thesis, we will define imaging as creating a visual representation - an image - of physical objects and scenes. Humans are able to perceive and interpret these images thanks to our visual system, that has evolved over millions of years. Developed from the simple light-sensitive cells of early organisms to the sophisticated eyes of modern humans, this system is providing us with the ability to see color, depth, or movement.

Over centuries, scholars and artists showed great interest in this process of creating visual representations to understand our own senses, as well as to be able to depict the world around us more accurately. The natural phenomenon of light passing through a small aperture, called camera obscura, helped us understand fundamental principles of optics and image formation, and according to some, greatly contributed to realism of European art during Middle Ages. We've been refining tools and techniques to enhance our vision, and improve everyday life. Devices like telescopes and microscopes allowed us to observe objects at different scales, unveiling worlds that would otherwise remain invisible to the naked eye. Eyeglasses and contact lenses correct refractive errors, demonstrating that our visual experiences can not only be externally altered, but also subjectively improved, for millions of people.

Photography emerged in the 19th century as a groundbreaking invention, offering a novel approach to image creation. Unlike paintings or drawings, which rely on an artist's interpretation, photography directly captures and permanently records the physical world using light-sensitive materials. Early photographic methods required long exposure times, often making it possible to photograph only still objects and posed portraits. The daguerreotype, which used a silver-coated copper plate, was one of the first widely adopted photographic processes. Later, more versatile film technologies significantly reduced exposure times and made photography more accessible to the wider public. This ability to easily and accurately record the physical world had significant consequences. Newspapers, now able to include actual photographs, found a powerful new way to share news and social commentary through photojournalism. On a personal level, photography allowed people to preserve their own histories through portraits and the documentation of special occasions.

Since the beginning of the 21st century, digital photography has become the dominant form of capturing and sharing images. Initially limited to standalone devices, digital cameras are now integrated into smartphones and personal computers. Electronic sensors capture and convert light into digital data, facilitating easy manipulation and image sharing, including

over the Internet. This digital nature has led to the popularity of standardized formats, ensuring compatibility across diverse platforms.

However, the ease of image creation and sharing, coupled with the vast amount of image data generated daily, may present unexpected challenges for storage and transmission. In 2024, an estimated 6 billion photos are taken daily, with 14 billion images shared on social media platforms. Especially in the context of thesis, the resulting volume of image data necessitates efficient delivery. Fortunately, optimization methods exist, relying on core concepts of image representation, encoding, and compression, to ensure efficient delivery across networks. We will explore these in the subsequent sections.

2.1.1 Digital imaging

Digital imaging, like traditional photography, relies on the fundamental principle of capturing light to create a visual representation. Most image acquisition methods, whether they involve cell phone cameras, DSLR (digital single-lens reflex) and mirrorless cameras, or even scanners, are variations of the classical optical camera. Concepts such as the relationship between aperture and shutter speed to achieve proper exposure are universal, regardless of whether the capture method is analog or digital.

However, digital imaging distinguishes itself by representing images as discrete, two-dimensional arrays of numerical data. This involves transforming light into a grid of distinct pixels. Each pixel may contain one or more channels — components that record different aspects of light, often corresponding to how our eyes perceive color. If only one channel is present, the image is effectively grayscale, as was the case with the first digital camera invented in 1975 by Steven Sasson.

To create a digital image, the camera must first capture the light that's focused onto its sensor. The position of the camera and its lens determine what portion of the scene is captured by each tiny pixel on the sensor — this is known as spatial sampling. The camera also controls how long it gathers light at each pixel, called temporal sampling, and is determined by the exposure time, or using photographic terminology, shutter speed. Thanks to the photovoltaic effect, the amount of the light captured can be converted into digital value through a process called quantization. The number of distinct values that can be represented for each pixel depends on the bit depth of the image.

Digital cameras primarily use two image sensor technologies: CCD (Charge-Coupled Device) and CMOS (Complementary Metal-Oxide-Semiconductor). Early digital cameras utilized CCD sensors, which offer high sensitivity and low noise due to their charge transfer method, where all charge is shifted across the chip to a single output node. However, this sequential readout makes them slower for continuous shooting or video capture. CCDs also tend to consume more power and are more prone to blooming, where overexposed pixels overflow. In contrast, CMOS sensors have seen rapid development since the 1990s. While initially lagging in sensitivity and noise performance comparing to CCDs, CMOS sensors have greatly improved through advancements like on-chip noise reduction. Their lower cost, lower power consumption, and faster readout speeds have led to CMOS sensors becoming more popular than CCDs and being found in a wide range of devices, from smartphones and webcams to DSLRs, mirrorless cameras, and even scientific imaging equipment.

To accurately capture color, most digital cameras use a color filter array (CFA) placed over the sensor. The most common type is the Bayer filter, named after its inventor, Bryce

Bayer, working at the time at Eastman Kodak. This filter arranges red, green, and blue filters in a specific pattern over the pixels. Because the human eye is more sensitive to green light, or to be specific, *because the green signal contributes more to the luminance signal than the red or blue signals* [], we can find twice as many green filters as red or blue - because of this property, such pattern is also known as RGGB. Cameras capture a full-color image by interpolating the missing color information for each pixel, in a process known as demosaicing.

Following the Bayer filter example, it is common for color images to have three channels, such as red, green, and blue (RGB), each with its own bit depth. If each channel in an RGB image has an 8-bit resolution, it allows for 256 different values per channel, resulting in over 16 million possible color combinations for each pixel. The resolution of an image refers to the total number of pixels it contains and is often measured in megapixels (millions of pixels). Higher resolution images have more pixels, what allows for larger prints and more detailed displays. The image resolution, along with the bit depth, directly affects the size of the uncompressed image file. However, because images often contain areas of similar color, compression techniques, discussed in the next section, are frequently employed to reduce file sizes, sometimes even during the image capture process within the camera.

2.1.2 Image data compression

2.1.3 Image formats and their usage

2.1.4 Raster and vector graphics

2.2 Cloud Computing

2.2.1 Cloud deployment models

2.2.2 Event-driven architectures

2.2.3 Cloud service benchmarking

2.2.4 Performance variability of serverless platforms

2.3 Image Optimization for the Web

2.3.1 Impact of images on browsing experience

2.3.2 Bandwidth usage, visual quality, and browser support evolution

2.3.3 Optimizing latency of image delivery

2.3.4 Deployment methods for cloud-based image optimization

3 System Design

3.1 Image processing workload design

3.1.1 Serverless environment selection

3.1.2 Image manipulation library selection

3.1.3 Input image and output image selection

3.2 Benchmark design

3.2.1 Serverless platform and underlying hardware

3.2.2 Time of execution

3.2.3 Output formats and image widths

3.2.4 Available CPU / memory

3.2.5 Cold starts

3.3 Workload generation and benchmark execution

3.3.1 Serverless benchmark deployment

3.3.2 Scheduling workload executions

3.3.3 Results collection and data analysis

4 Results

4.1 Performance metrics

4.1.1 Cold start duration

4.1.2 Image processing duration

4.1.3 Memory utilization

4.2 Performance variability

4.2.1 General variability in performance metrics

4.2.2 Temporal variability

4.2.3 By image format

4.2.4 By platform configuration

5 Evaluation

5.1 Performance variability analysis

5.2 User vs operator perspective

5.3 Cloud computing as a utility

6 Related Work

6.1 Serverless performance research

6.2 Image processing research

7 Conclusion